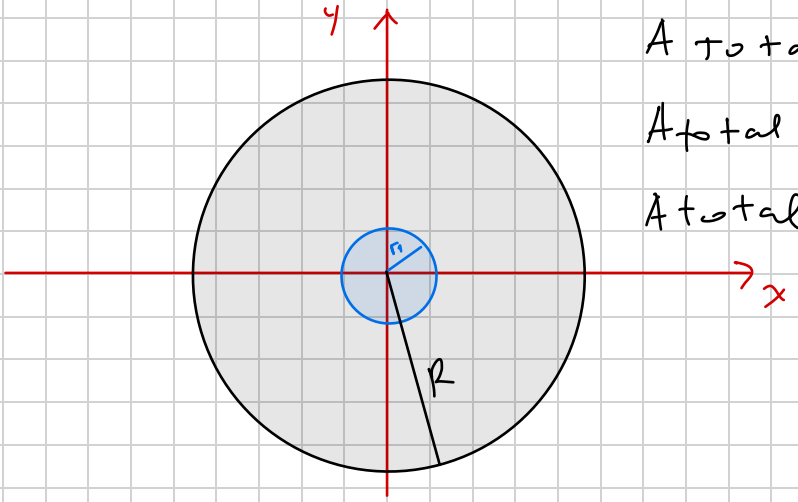


$$A_{total} = A_R - A_{r_1}$$

$$A_{total} = \pi R^2 - \pi r_1^2$$

$$A_{total} = \pi (R^2 - r_1^2)$$

$$x^2 + y^2 = R^2$$

$$y = \pm\sqrt{R^2 - x^2}$$

$$A_R = 2\int_0^R 2\sqrt{R^2 - x^2}\,dx = 4\int_0^R \sqrt{R^2 - x^2}\,dx$$

Sustitución Trigonométrica

$$x = R\,\text{Seno}$$
$$dx = R\cos\theta\,d\theta$$

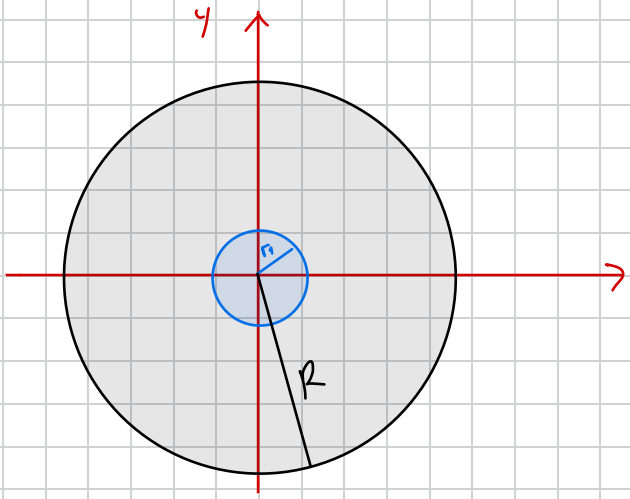
$$x^2 + y^2 = r_1^2$$

$$y = \pm\sqrt{r_1^2 - x^2}$$

$$A_{r_1} = 2\int_0^{r_1} 2\sqrt{r_1^2 - x^2}\,dx = 4\int_0^{r_1} \sqrt{r_1^2 - x^2}\,dx$$

Sustitución Trigonométrica

$$x = R\,\text{Seno}$$
$$dx = R\cos\theta\,d\theta$$



$$A = \iint_D dy\,dx = \iint_D dx\,dy$$

$$A_R = 4\int_0^R \int_0^{\sqrt{R^2-x^2}} dy\,dx = 4\int_0^R \sqrt{R^2-x^2}\,dx$$

$x = R\,\text{sen}\,\theta$
$dx = R\cos\theta\,d\theta$

$$= 4\int_0^{\pi/2} \sqrt{R^2 - R^2\text{sen}^2\theta}\,(R\cos\theta)\,d\theta$$

$$R^2(1-\text{sen}^2\theta)$$
$$R^2\cos^2\theta$$

$$\text{sen}\,\theta = \frac{x}{R}$$

$$\theta = \text{sen}^{-1}\left(\frac{x}{R}\right)$$

$x = 0$	$x = R$
$\theta = 0$	$\theta = \pi/2$

$$= 4\int_0^{\pi/2} R\cos\theta\, R\cos\theta\,d\theta$$

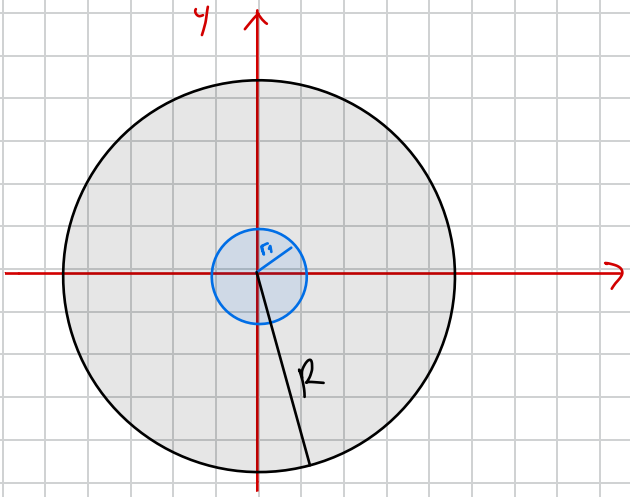
$$= 4R^2\int_0^{\pi/2} \cos^2\theta\,d\theta$$

$$= 4R^2\int_0^{\pi/2} \frac{1+\cos 2\theta}{2}\,d\theta$$

$$= 2R^2\int_0^{\pi/2} (1+\cos 2\theta)\,d\theta$$

$$= 2R^2\left(\theta + \frac{\text{Sen}(2\theta)}{2}\right)\Big|_0^{\pi/2}$$

$$= 2R^2\left(\pi/2\right) = \pi R^2$$



$$A = \iint_D dy\,dx = \iint_D dx\,dy$$

$$A_R = 4\int_0^{r_1}\int_0^{\sqrt{r_1^2-x^2}} dy\,dx = 4\int_0^{r_1}\sqrt{r_1^2-x^2}\,dx$$

$x = r_1 \operatorname{sen}\theta$
$dx = r_1\cos\theta\,d\theta$

$$= 4\int_0^{\pi/2}\sqrt{r_1^2 - {}^2\operatorname{sen}^2\theta}\;(r_1\cos\theta)\,d\theta$$

$$^2(1-\operatorname{sen}^2\theta)$$
$$^2\cos^2\theta$$

$$\operatorname{sen}\theta = \frac{x}{r_1}$$

$$\theta = \operatorname{sen}^{-1}\left(\frac{x}{r_1}\right)$$

$x = 0$	$x = r_1$
$\theta = 0$	$\theta = \pi/2$

$$= 4\int_0^{\pi/2} r_1\cos\theta\, r_1\cos\theta\,d\theta$$

$$= 4r_1^2\int_0^{\pi/2}\cos^2\theta\,d\theta$$

$$= 4r_1^2\int_0^{\pi/2}\frac{1+\cos 2\theta}{2}\,d\theta$$

$$= 2r_1^2\int_0^{\pi/2}(1+\cos 2\theta)\,d\theta$$

$$= 2r_1^2\left(\theta + \frac{\operatorname{Sen}(2\theta)}{2}\right)\Bigg|_0^{\pi/2}$$

$$= 2r_1^2\left(\pi/2\right) = \pi r_1^2$$

Marco de referencia, coordenadas polares.

Cartesianas $\longrightarrow$ polares $\quad 0 \le \theta \le 2\pi$

$(x,y) \longrightarrow (r,\theta)$ $\quad \dfrac{\partial(x,y)}{\partial(r,\theta)}$

$x = r\cos\theta$

$y = r\operatorname{sen}\theta$

$$\iint_D f(x,y)\underbrace{dA}_{\substack{dy\,dx \\ dx\,dy}} = \iint_{D_{(r,\theta)}} f(r,\theta)\,|J|\,dA \qquad dA = dr\,d\theta$$

$$J = \begin{bmatrix} \dfrac{\partial x}{\partial r} & \dfrac{\partial x}{\partial \theta} \\ \dfrac{\partial y}{\partial r} & \dfrac{\partial y}{\partial \theta} \end{bmatrix} = \begin{bmatrix} \cos\theta & -r\operatorname{sen}\theta \\ \operatorname{sen}\theta & r\cos\theta \end{bmatrix}$$

$$|J| = r\cos^2\theta + r\operatorname{sen}^2\theta = r(\cos^2\theta + \operatorname{sen}^2\theta) = r$$

$$= \iint_{D_{(r,\theta)}} f(r,\theta)\,r\,dr\,d\theta$$

$$A = 4\int_0^{\pi/2}\int_{r_1}^{R} r\,dr\,d\theta = 4\int_0^{\pi/2} \frac{r^2}{2}\Big|_{r_1}^{R} d\theta$$

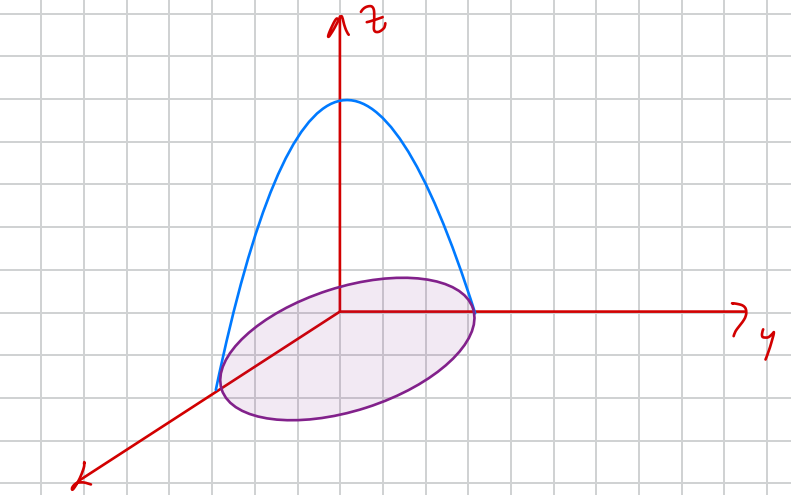
$$= 2\int_0^{\pi/2} (R^2 - r_1^2)\,d\theta$$

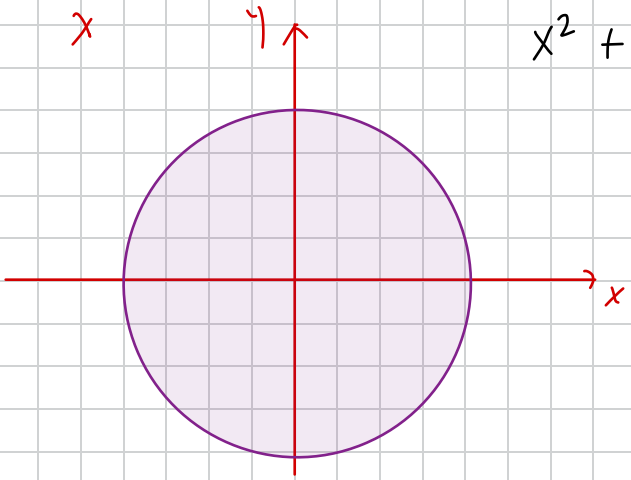
$$= 2(R^2 - r_1^2)\,\theta\Big|_0^{\pi/2}$$

$$= 2(R^2 - r_1^2)\,\frac{\pi}{2}$$

$$= \pi(R^2 - r_1^2)$$

Ejemplo: calcular el volumen V del sólido acotado por el paraboloide $z = 4 - x^2 - y^2$ y por el plano xy.





$x^2 + y^2 = 4$

Integral en coordenadas cartesianas:

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ii) cálculo de volumen: $V = \int_a^b \int_{g(x)}^{f(x)} f(x,y)\, dx\, dy$

$z = f(x,y)$

$$V = 4 \int_0^2 \int_0^{\sqrt{4-x^2}} (4 - x^2 - y^2)\, dy\, dx$$

Integral en coordenadas Polares.

$$V = 4 \int_0^{\pi/2} \int_0^2 (4 - r^2)\, r\, dr\, d\theta$$

$$V = 4\int_0^{\pi/2}\int_0^{2}(4 - r^2)\, r\, dr\, d\theta$$

$$V = 4\int_0^{\pi/2}\int_0^{2}(4r - r^3)\, dr\, d\theta$$

$$V = 4\int_0^{\pi/2}\left(2r^2 - \frac{r^4}{4}\right)\Big|_0^{2} d\theta$$

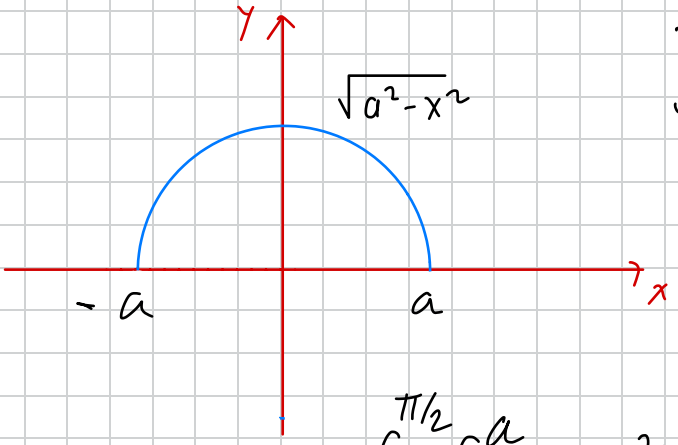
$$V = 4\,(8 - 4)\int_0^{\pi/2} d\theta$$

$$V = 16\,\theta\Big|_0^{\pi/2}$$

$$V = 16 \cdot \frac{\pi}{2} = \boxed{8\pi\, u^3}$$

Ejemplo: Evaluar la integral:

$$\int_{-a}^{a}\int_{0}^{\sqrt{a^2-x^2}} e^{-(x^2+y^2)}\,dy\,dx$$



$x = r\cos\theta$

$y = r\operatorname{sen}\theta$

$u = r^2$

$du = 2r\,dr$

$\frac{du}{2} = dr$

$$= 2\int_{0}^{\pi/2}\int_{0}^{a} e^{-r^2} r\,dr\,d\theta$$

$$= 2\int_{0}^{\pi/2}\int \frac{1}{2} e^{-u}\,du\,d\theta$$

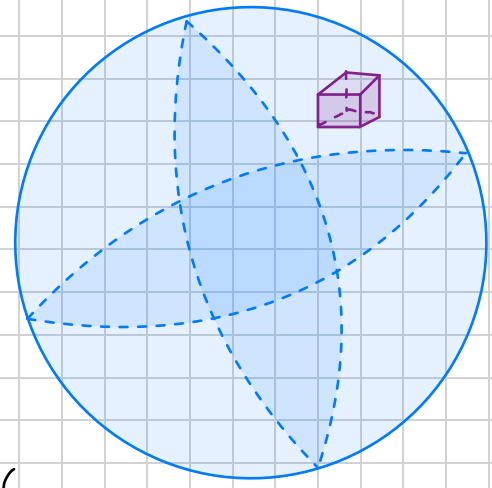
$$= -\int_{0}^{\pi/2} e^{-r^2}\Big|_{0}^{a}\,d\theta$$

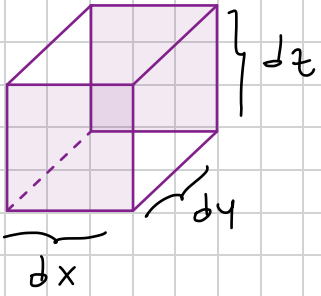
$$= -(e^{-a^2}-1)\int_{0}^{\pi/2} d\theta$$

$$= (1-e^{-a^2})\,\theta\Big|_{0}^{\pi/2}$$

$$= \frac{\pi}{2}(1-e^{-a^2}).$$

Integrales triples





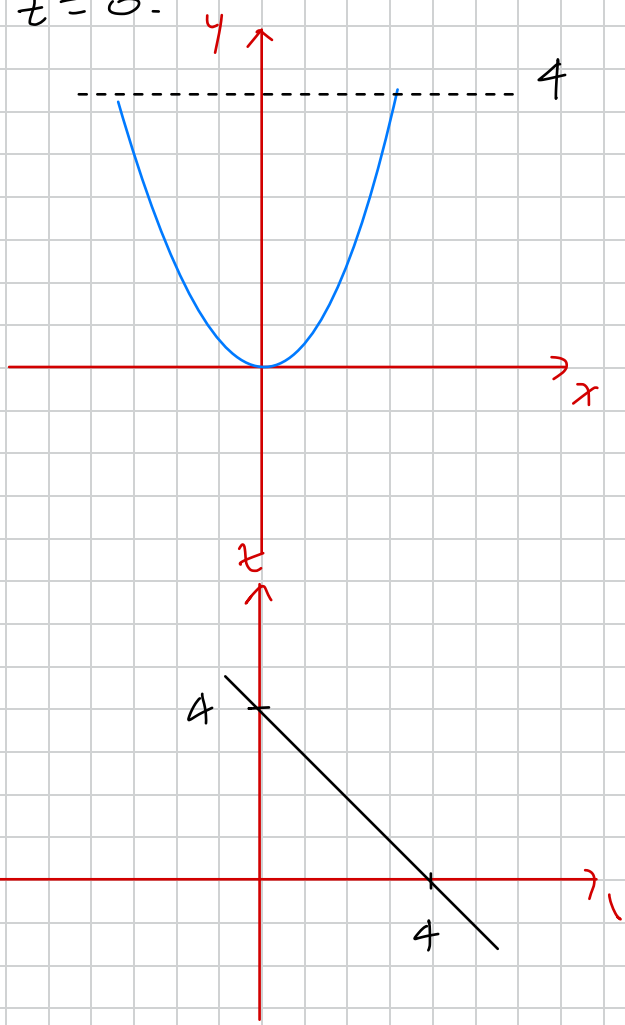
$$\int dV = \iiint dz\,dy\,dx$$

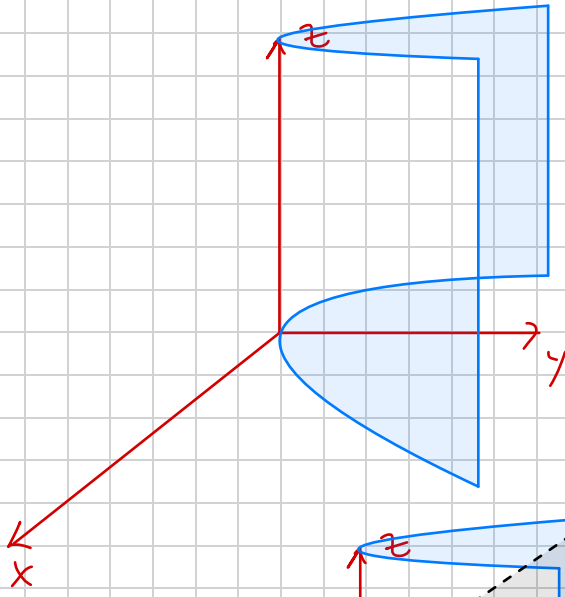
$$V = \int_a^b \int_{h_1(x)}^{h_2(x)} \int_{K_1(x,y)}^{K_2(x,y)} dz\,dy\,dx$$

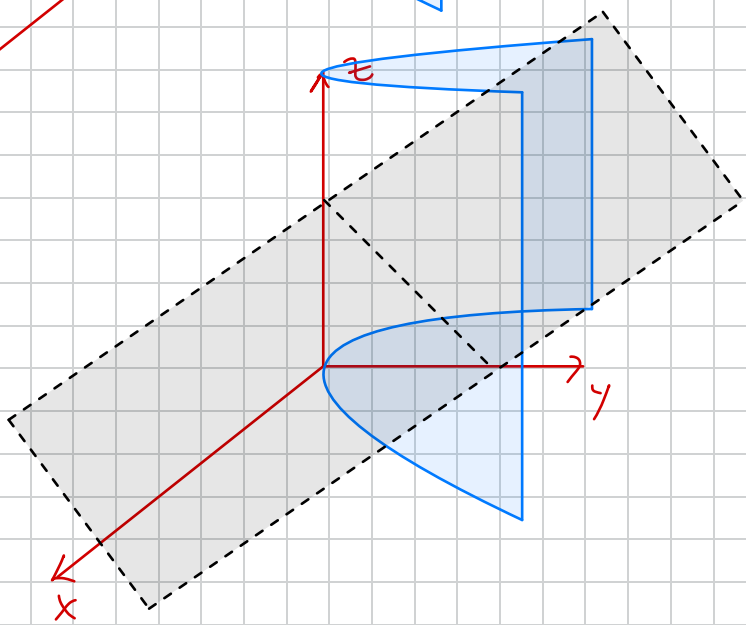
$$\iiint_Q f(x,y,z)\,dV = \int_a^b \int_{h_1(x)}^{h_2(x)} \int_{K_1(x,y)}^{K_2(x,y)} f(x,y,z)\,dz\,dy\,dx$$

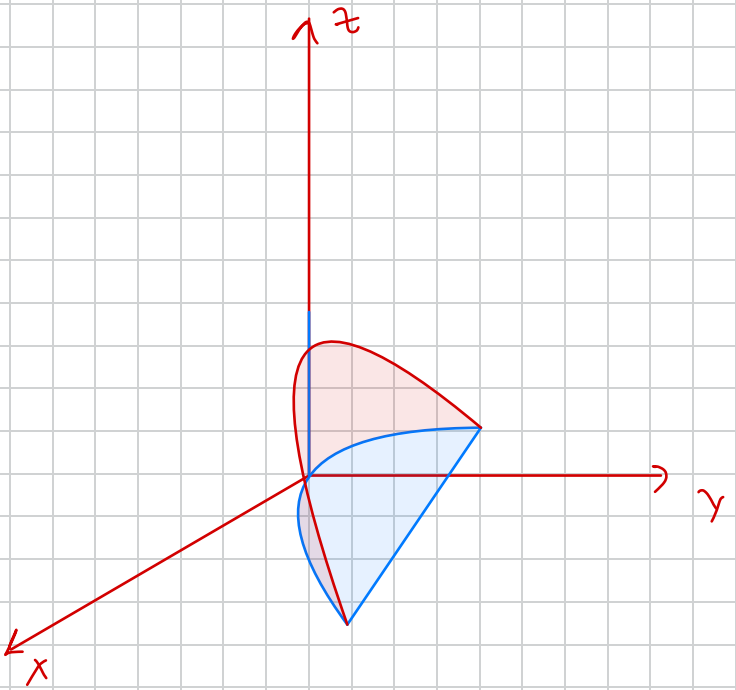
Ejemplo: Calcular el volumen del sólido acotado por el cilindro $y = x^2$ y los planos $y + z = 4$ y $z = 0$.

Hoja parabólica.









$$V = \int_0^4 \int_{-\sqrt{y}}^{\sqrt{y}} \int_0^{4-y} dz\,dx\,dy$$

$$V = \int_0^4 \int_{-\sqrt{y}}^{\sqrt{y}} (4-y)\,dx\,dy$$

$$V = \int_0^4 2(4-y)\sqrt{y}\,dy$$

$$V = 2\int_0^4 (4y^{1/2} - y^{3/2})\,dy$$

$$V = 2\left(\frac{8y^{3/2}}{3} - \frac{2y^{5/2}}{5}\right)\Big|_0^4$$

$$V = 2\left(\frac{8}{3}(4)^{3/2} - \frac{2}{5}(4)^{5/2}\right) = \frac{256}{15}\,ud^3$$

$$V = \int_{-2}^{2} \int_{x^2}^{4} \int_0^{4-y} dz\,dy\,dx = \int_{-2}^{2} \int_{x^2}^{4} (4-y)\,dy\,dx$$

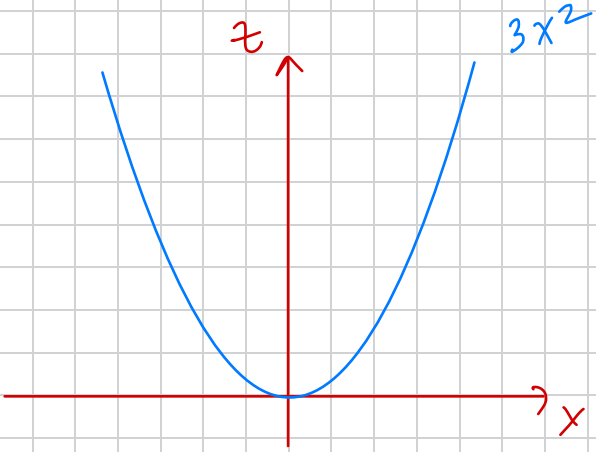
$$V = \int_{-2}^{2} \left(4y - \frac{y^2}{2}\right)\Big|_{x^2}^{4} dx = \int_{-2}^{2} \left(8 - 4x^2 + \frac{x^4}{2}\right) dx$$

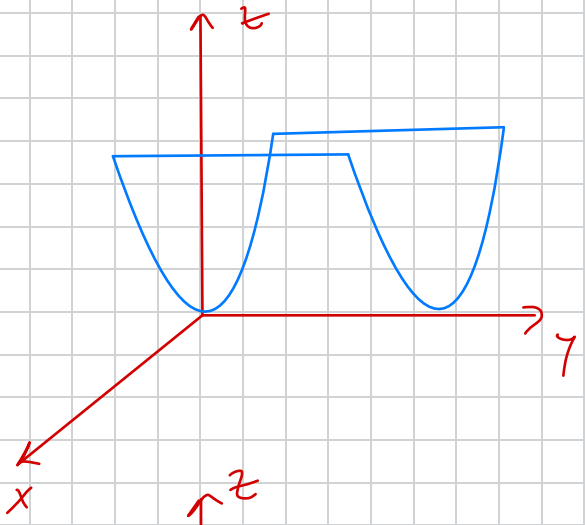
$$V = \left(8x - \frac{4}{3}x^3 + \frac{x^5}{10}\right)\Big|_{-2}^{2}$$

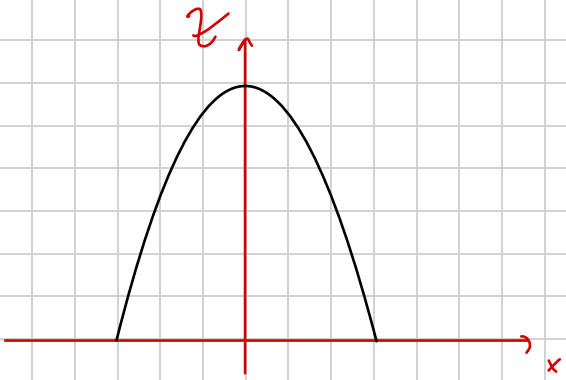
$$V = \left(16 - \frac{4}{3}(8) + \frac{32}{10}\right) - \left(8(-2) - \frac{4}{3}(-2)^3 + \frac{(-2)^5}{10}\right)$$

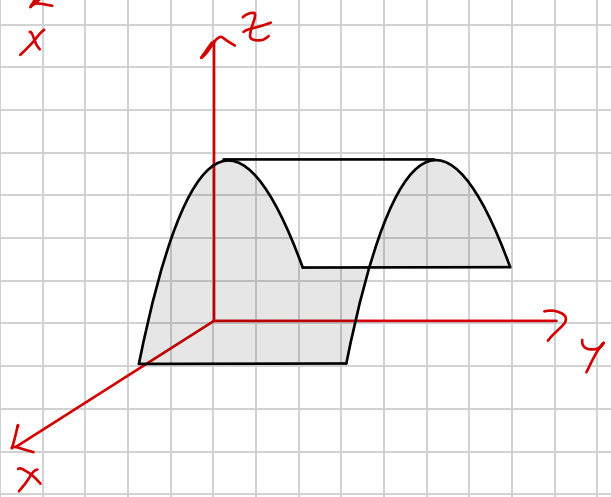
$$V = \frac{256}{15}\,ud^3$$

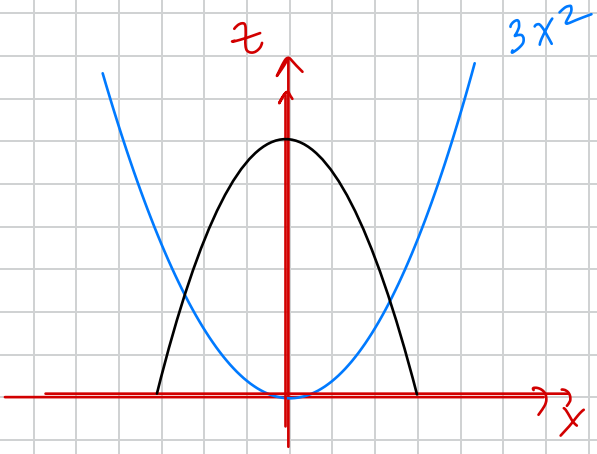
Ejemplo: calcular el volumen de la región Q acotada por las gráficas $z = 3x^2$; $z = 4 - x^2$; $y = 0$ y $z + y = 6$.

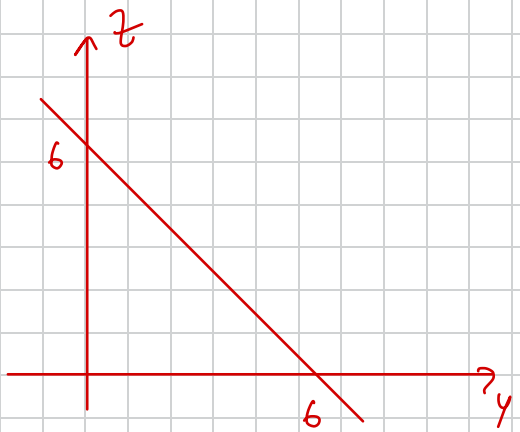




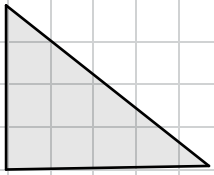












$$V = \int_{-1}^{1} \int_{3x^2}^{4-x^2} \int_{0}^{6-z} dy\,dz\,dx$$

$$4 - x^2 = 3x^2$$

$$4 = 4x^2$$

$$1 = x^2$$

$$x^2 - 1 = 0$$

$$(x-1)(x+1) = 0$$

$$V = \int\int (6-z)\,dz\,dx$$

$$V = \int_{-1}^{1} \left(6z - \frac{z^2}{2}\right)\Big|_{3x^2}^{4-x^2} dx$$

$$= \int_{-1}^{1} \left(24 - 6x^2 - 18x^2 - \frac{1}{2}\left((4-x^2)^2 - 9x^4\right)\right) dx$$

$$24 - 6x^2 - \frac{1}{2}(16 - 8x^2 + x^4 - 9x^4) - 18x^2$$

$$24 - 6x^2 - \frac{1}{2}(16 - 8x^2 - 8x^4) - 18x^2$$

$$24 - 6x^2 - 8 + 4x^2 + 4x^4 - 18x^2$$

$$V = \int_{-1}^{1} (16 - 20x^2 + 4x^4)\,dx$$

$$= \left(16x - \frac{20}{3}x^3 + \frac{4}{5}x^5\right)\Big|_{-1}^{1}$$

$$\left(16 - \frac{20}{3} + \frac{4}{5}\right) - \left(-16 + \frac{20}{3} - \frac{4}{5}\right) =$$

$$16 + 16 - \frac{20}{3} - \frac{20}{3} + \frac{4}{5} + \frac{4}{5}$$

$$32 - \frac{40}{3} + \frac{8}{5} = \frac{304}{15}\ ud^3$$